Task 7 Report

Abaqus Wind Loading on Cooling Towers

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Contents

Chapter 1

Objective

This Task 7 study expands the previous three-tower screening into a full structural comparison across the eight Task 6 scenarios and the three optimization algorithms SA, PSO, and BFGS.

The engineering objective is to identify which Task 6 concepts remain convincing once the towers are checked under self-weight, one-direction wind loading, and linear buckling in Abaqus.

Chapter 2

Selection Rule and Scenario Matrix

The Task 7 matrix contains **24 refined presentation models**. For scenarios S1 to S7, the preferred choice is the lowest-area engineering-feasible Task 6 run for each scenario/algorithm pair. If no engineering-feasible run exists, the workflow falls back to the lowest-area mathematically compliant run. If neither exists, the best penalized run is kept only as a warning case. Scenario S8 is intentionally carried as warning-only because it has no compliant Task 6 solution in the current study.

Across the 24 selected cases, there are **16 engineering-feasible selections**, **3 mathematical fallbacks**, and **5 warning-only cases**.

Case	Status	Selection basis	Task 6 run	Decision mode	Task 6 area (m^2)
S1 / BFGS	Warning case	best_penalized_warning	6	radii	7782.01
S1 / PSO	Engineering-feasible	lowest_area_engineering_feasible	7	radii	7804.56
S1 / SA	Engineering-feasible	lowest_area_engineering_feasible	5	radii	7780.73
S2 / BFGS	Engineering-feasible	lowest_area_engineering_feasible	2	radii	7741.67
S2 / PSO	Math fallback	lowest_area_mathematical_fallback	0	radii	9075.03
S2 / SA	Engineering-feasible	lowest_area_engineering_feasible	5	radii	7744.01
S3 / BFGS	Engineering-feasible	lowest_area_engineering_feasible	6	radii	7741.69
S3 / PSO	Engineering-feasible	lowest_area_engineering_feasible	6	radii	7743.34
S3 / SA	Engineering-feasible	lowest_area_engineering_feasible	6	radii	7742.77
S4 / BFGS	Engineering-feasible	lowest_area_engineering_feasible	1	radii	7779.85
S4 / PSO	Warning case	best_penalized_warning	1	radii	7931.66
S4 / SA	Engineering-feasible	lowest_area_engineering_feasible	7	radii	7769.88
S5 / BFGS	Engineering-feasible	lowest_area_engineering_feasible	0	heights	7742.56
S5 / PSO	Engineering-feasible	lowest_area_engineering_feasible	6	heights	7747.93
S5 / SA	Engineering-feasible	lowest_area_engineering_feasible	6	heights	7749.52
S6 / BFGS	Engineering-feasible	lowest_area_engineering_feasible	1	heights	7741.72
S6 / PSO	Engineering-feasible	lowest_area_engineering_feasible	3	heights	7748.25
S6 / SA	Engineering-feasible	lowest_area_engineering_feasible	0	heights	7783.24
S7 / BFGS	Engineering-feasible	lowest_area_engineering_feasible	1	joint	7740.66
S7 / PSO	Math fallback	lowest_area_mathematical_fallback	9	joint	8324.25
S7 / SA	Math fallback	lowest_area_mathematical_fallback	9	joint	7890.85
S8 / BFGS	Warning case	best_penalized_warning	5	joint	7813.24
S8 / PSO	Warning case	best_penalized_warning	8	joint	7899.07
S8 / SA	Warning case	best_penalized_warning	0	joint	7718.27

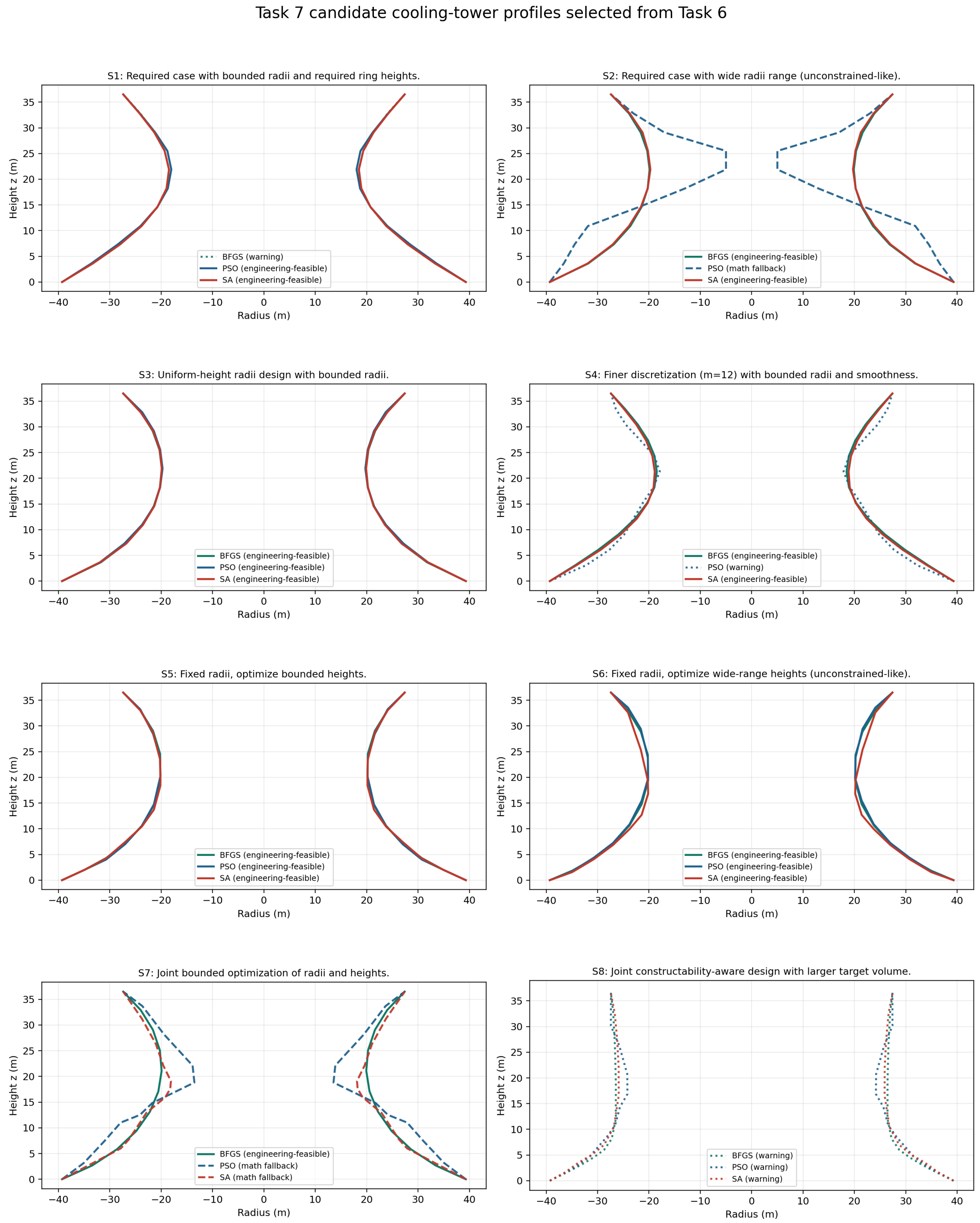

Figure 2.1: Task 7 candidate profiles

Chapter 3

Material, Loading, and CAE Setup

The structural baseline keeps the same assumptions for every model: equivalent reinforced concrete with $E = 33GPa$, $nu = 0.20$, and `rho = 2500 kg/m^3`, together with a constant shell thickness of `0.20 m`.

The reference wind speed is `30.0 m/s`, which gives a dynamic pressure of `562.5 Pa` through $q = 0.5rhoV^2$.

Wind loading is interpreted as one-direction external action: windward sectors receive positive external pressure and leeward sectors receive suction. This is not internal cabin-style pressurization.

The directional pressure model is:

$$C_p(\theta) = \mathrm{clip}(0.8\cos\theta, -0.5, 0.8), \quad p(\theta) = q\,C_p(\theta)$$

Task 7 uses a strict `Y`-up coordinate convention: `Y` is vertical, the horizontal wind plane is `X/Z`, and self-weight acts along `-Y`.

The clean user-facing Abaqus/CAE database contains the **24 refined models only**, with one gravity load and one wind load per model. The automated solver decks still use the same circumferential pressure law in sectorized form for coarse-versus-refined verification.

The refined study uses one uniform high mesh policy for all 24 towers (`40` circumferential divisions, `2` axial subdivisions per segment). Refined node counts range from `840` to `1000` nodes, which keeps every case within the Abaqus Learning Edition limit while using the highest common density.

Solver-deck mesh compliance (`mesh_summary.csv`) is tracked separately from CAE integrity: refined solver decks span `840` to `1000` nodes and remain within the Abaqus LE cap.

CAE-native integrity (`cae_integrity_audit.json`) reports a model-node range of `840` to `1075`. This audit validates model-tree integrity and CAE consistency, not LE solve-cap compliance.

A deterministic input-deck audit runs before solving and confirms that all 48 decks use identical material constants, gravity definition, and wind-pressure sector logic.

The towers look squat in Abaqus because the geometry is genuinely squat: the Task 6 cooling towers are about `36.5 m` tall for a base diameter of about `78.6 m`, so the aspect ratio is below one. This is source geometry, not an Abaqus distortion.

Chapter 4

Geometry Integrity Check

The Task 7 generator preserves the exact Task 6 meridian coordinates. A geometry verification pass is written before solving and checks the mesh rings against the selected Task 6 source profile for every coarse and refined job.

Mesh	**Max total-height diff (m)**	**Max ring-radius diff (m)**	**Max ring-z diff (m)**
coarse	0.000000e+00	0.000000e+00	0.000000e+00
refined	0.000000e+00	0.000000e+00	0.000000e+00

Chapter 5

Refined Structural Comparison Across All 24 Cases

Case	Status	Task 6 area (m^2)	Max disp. (mm)	Max stress (MPa)	Buckling factor
S1 / BFGS	Warning case	7782.01	4.264	2.262	20.907
S1 / PSO	Engineering-feasible	7804.56	4.126	2.244	21.300
S1 / SA	Engineering-feasible	7780.73	4.627	2.368	21.481
S2 / BFGS	Engineering-feasible	7741.67	10.202	3.867	23.563
S2 / PSO	Math fallback	9075.03	94.040	10.268	-3.229
S2 / SA	Engineering-feasible	7744.01	9.968	3.818	24.549
S3 / BFGS	Engineering-feasible	7741.69	10.153	3.834	23.529
S3 / PSO	Engineering-feasible	7743.34	9.850	3.794	23.496
S3 / SA	Engineering-feasible	7742.77	9.036	3.526	23.834
S4 / BFGS	Engineering-feasible	7779.85	3.831	2.282	20.267
S4 / PSO	Warning case	7931.66	12.073	4.541	22.491
S4 / SA	Engineering-feasible	7769.88	4.287	2.296	19.753
S5 / BFGS	Engineering-feasible	7742.56	8.695	4.148	22.725
S5 / PSO	Engineering-feasible	7747.93	11.481	4.941	19.566
S5 / SA	Engineering-feasible	7749.52	10.229	4.642	23.232
S6 / BFGS	Engineering-feasible	7741.72	10.522	5.312	24.772
S6 / PSO	Engineering-feasible	7748.25	11.907	6.705	21.611
S6 / SA	Engineering-feasible	7783.24	14.270	7.690	19.128
S7 / BFGS	Engineering-feasible	7740.66	10.105	4.134	24.948
S7 / PSO	Math fallback	8324.25	28.467	9.258	23.712
S7 / SA	Math fallback	7890.85	12.541	6.711	26.587
S8 / BFGS	Warning case	7813.24	10.319	5.446	19.273
S8 / PSO	Warning case	7899.07	9.235	4.493	27.335
S8 / SA	Warning case	7718.27	8.163	4.955	23.837

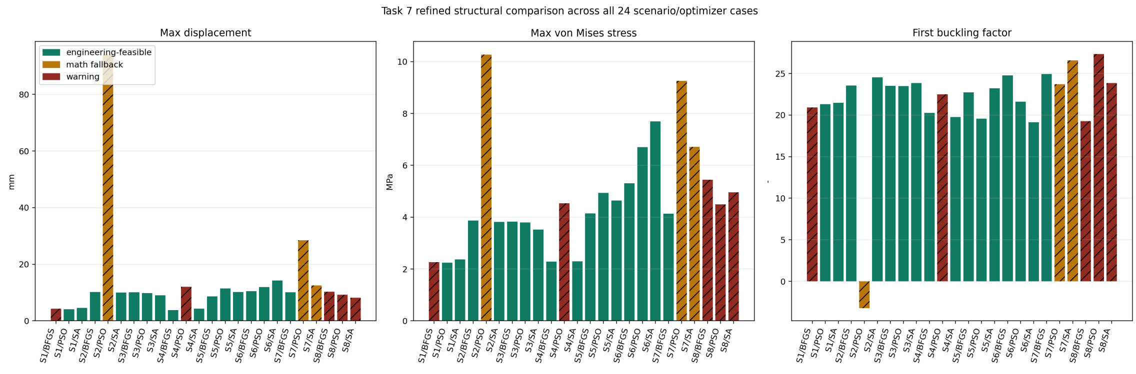

Figure 5.1: Task 7 refined comparison metrics

Chapter 6

Methodological Clarity and Global Ranking

6.1 Methodological Clarity

The global ranking is a rank-based weighted score on the refined results, designed to stay robust when one or two cases have extreme values.

For each metric, rank `1` is best and rank `24` is worst, with deterministic tie-breaks by scenario/algorithm identifiers.

$$R_{k,i} = textrankofcaseitextonmetrick$$

$$S_i = 0.45R_{mathrmbuckling,i} + 0.25R_{mathrmdisp,i} + 0.20R_{mathrmstress,i} + 0.10R_{mathrmarea,i}$$

$$J_i = S_i + P_i, quadP_iin0, 10, 20$$

Penalty points are `0` for engineering-feasible, `10` for mathematical fallback, and `20` for warning cases. The best model is the one with the lowest J_i.

6.2 Global Weighted Top-3

Rank	Case	Status	Weighted score	Penalty	Buckling factor	Max disp. (mm)	Max stress (MPa)	Area (m^2)
1	S7 / BFGS	Engineering-feasible	6.750	0.0	24.948	10.105	4.134	7740.66
2	S3 / SA	Engineering-feasible	7.050	0.0	23.834	9.036	3.526	7742.77
3	S2 / SA	Engineering-feasible	7.500	0.0	24.549	9.968	3.818	7744.01

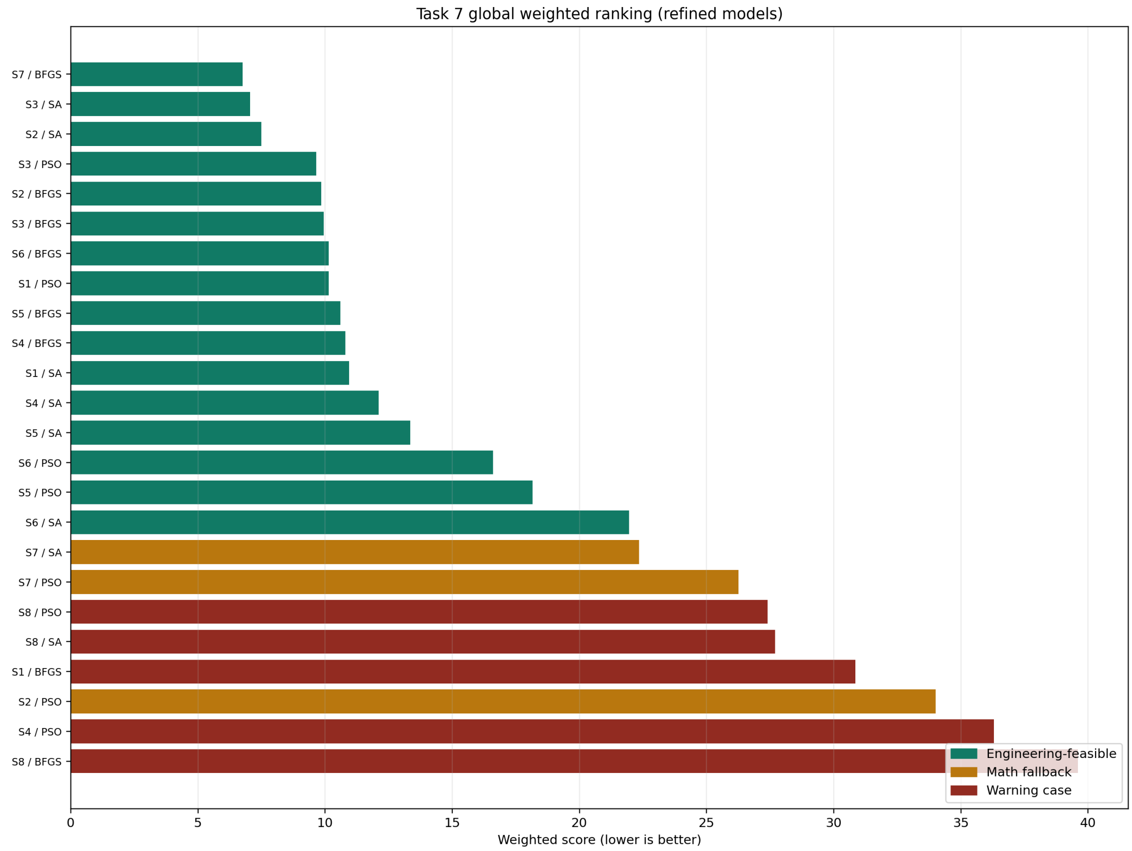

Figure 6.1: Task 7 weighted ranking

6.3 Top-3 by Criterion

6.3.1 Raw Top-3 (all statuses)

This raw criterion table keeps all statuses visible, so warning and fallback cases can appear if they are numerically extreme on a single indicator.

Criterion	Rank	Case	Status	Value
Buckling factor	1	S8 / PSO	Warning case	27.335
Buckling factor	2	S7 / SA	Math fallback	26.587
Buckling factor	3	S7 / BFGS	Engineering-feasible	24.948
Max displacement	1	S4 / BFGS	Engineering-feasible	3.831 mm
Max displacement	2	S1 / PSO	Engineering-feasible	4.126 mm
Max displacement	3	S1 / BFGS	Warning case	4.264 mm
Max stress	1	S1 / PSO	Engineering-feasible	2.244 MPa
Max stress	2	S1 / BFGS	Warning case	2.262 MPa
Max stress	3	S4 / BFGS	Engineering-feasible	2.282 MPa
Task 6 area	1	S8 / SA	Warning case	7718.27 m^2
Task 6 area	2	S7 / BFGS	Engineering-feasible	7740.66 m^2
Task 6 area	3	S2 / BFGS	Engineering-feasible	7741.67 m^2

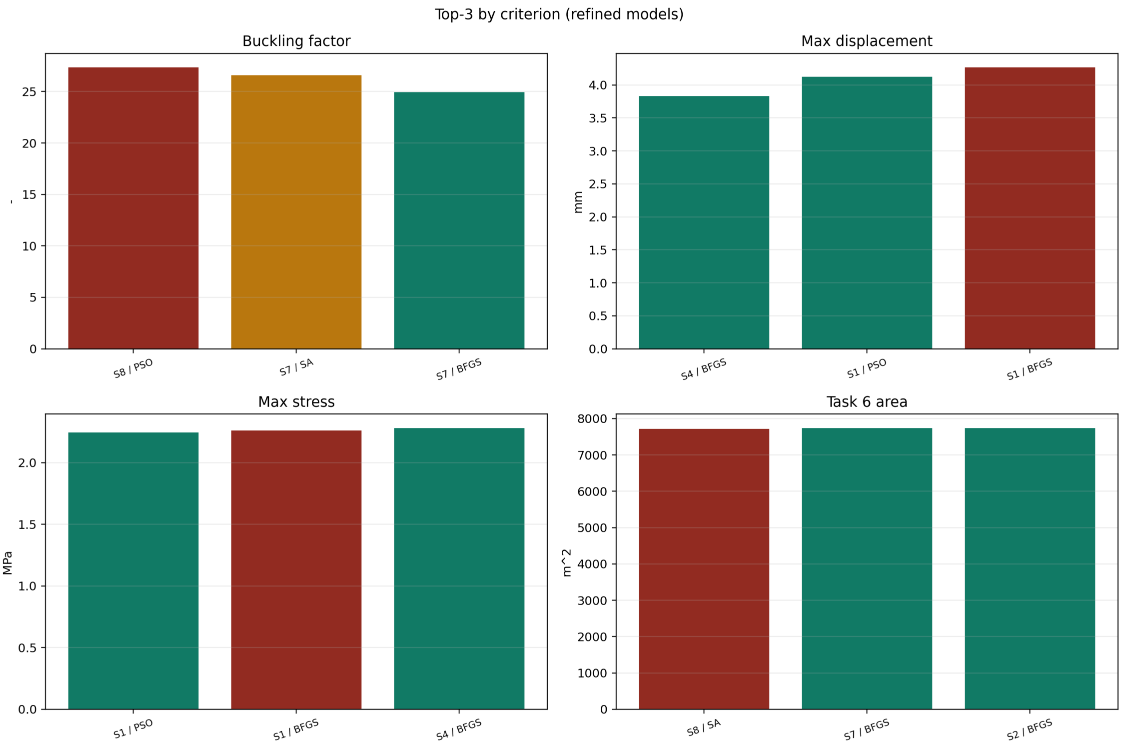

Figure 6.2: Task 7 criterion leaders

6.3.2 Engineering-eligible Top-3

This decision-grade table filters to engineering-feasible entries only.

Criterion	Rank	Case	Status	Value
Buckling factor	1	S7 / BFGS	Engineering-feasible	24.948
Buckling factor	2	S6 / BFGS	Engineering-feasible	24.772
Buckling factor	3	S2 / SA	Engineering-feasible	24.549
Max displacement	1	S4 / BFGS	Engineering-feasible	3.831 mm
Max displacement	2	S1 / PSO	Engineering-feasible	4.126 mm
Max displacement	3	S4 / SA	Engineering-feasible	4.287 mm
Max stress	1	S1 / PSO	Engineering-feasible	2.244 MPa
Max stress	2	S4 / BFGS	Engineering-feasible	2.282 MPa
Max stress	3	S4 / SA	Engineering-feasible	2.296 MPa
Task 6 area	1	S7 / BFGS	Engineering-feasible	7740.66 m^2
Task 6 area	2	S2 / BFGS	Engineering-feasible	7741.67 m^2
Task 6 area	3	S3 / BFGS	Engineering-feasible	7741.69 m^2

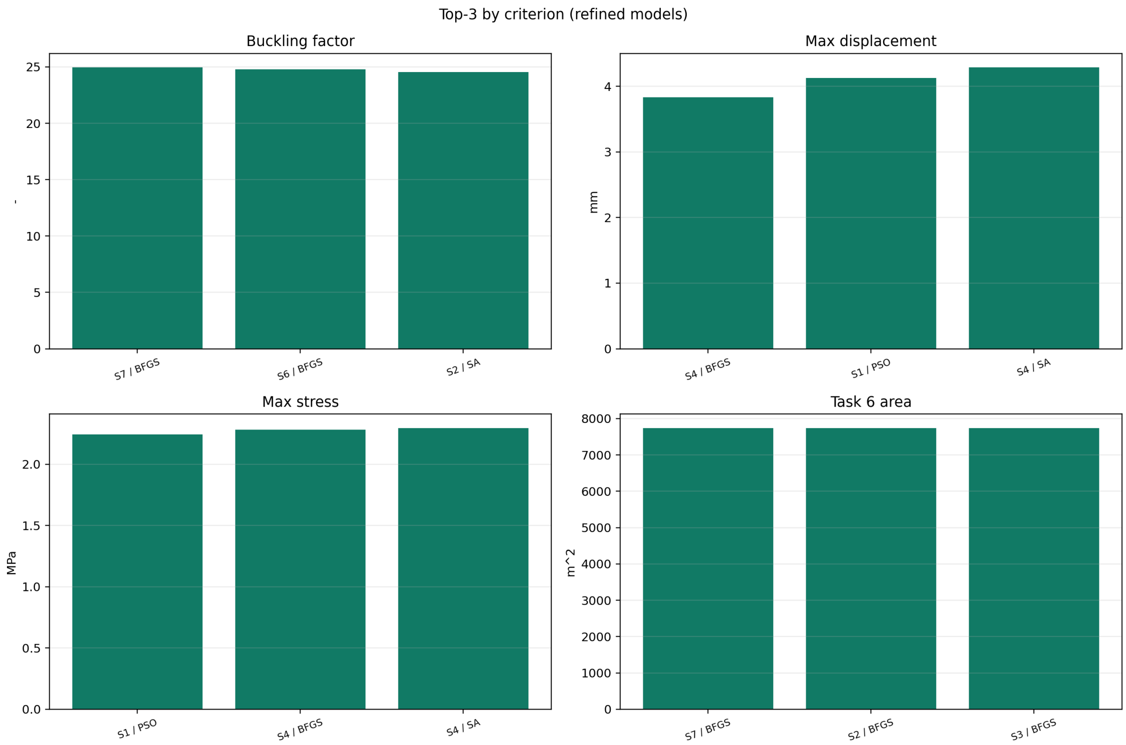

Figure 6.3: Task 7 engineering criterion leaders

Chapter 7

Scenario-by-Scenario Comparison

7.1 S1

Algorithm	Status	Weighted score	Task 6 area (m^2)	Max disp. (mm)	Max stress (MPa)	Buckling factor
BFGS	Warning case	30.850	7782.01	4.264	2.262	20.907
PSO	Engineering-feasible	10.150	7804.56	4.126	2.244	21.300
SA	Engineering-feasible	10.950	7780.73	4.627	2.368	21.481

Winner: S1 / PSO with weighted score `10.150`.

The score gap to second place (S1 / SA) is `0.800` points.

Spread/Risk: displacement spans `4.126` to `4.627` mm, stress spans `2.244` to `2.368` MPa, and buckling spans `20.907` to `21.481`.

Status caveat: warning-only entries are present for BFGS and are not decision-grade designs.

Critical note: scenario fairness is reduced by mixed compliance statuses; ranking penalties keep the comparison visible but do not make warning/fallback cases equivalent to engineering-feasible designs.

7.2 S2

Algorithm	Status	Weighted score	Task 6 area (m^2)	Max disp. (mm)	Max stress (MPa)	Buckling factor
BFGS	Engineering-feasible	9.850	7741.67	10.202	3.867	23.563
PSO	Math fallback	34.000	9075.03	94.040	10.268	-3.229
SA	Engineering-feasible	7.500	7744.01	9.968	3.818	24.549

Winner: S2 / SA with weighted score `7.500`.

The score gap to second place (S2 / BFGS) is `2.350` points.

Spread/Risk: displacement spans `9.968` to `94.040` mm, stress spans `3.818` to `10.268` MPa, and buckling spans `-3.229` to `24.549`.

Status caveat: mathematical fallback entries are present for PSO.

Critical note: scenario fairness is reduced by mixed compliance statuses; ranking penalties keep the comparison visible but do not make warning/fallback cases equivalent to engineering-feasible designs.

7.3 S3

Algorithm	Status	Weighted score	Task 6 area (m^2)	Max disp. (mm)	Max stress (MPa)	Buckling factor
BFGS	Engineering-feasible	9.950	7741.69	10.153	3.834	23.529
PSO	Engineering-feasible	9.650	7743.34	9.850	3.794	23.496
SA	Engineering-feasible	7.050	7742.77	9.036	3.526	23.834

Winner: S3 / SA with weighted score `7.050`.

The score gap to second place (S3 / PSO) is `2.600` points.

Spread/Risk: displacement spans `9.036` to `10.153` mm, stress spans `3.526` to `3.834` MPa, and buckling spans `23.496` to `23.834`.

Status caveat: all three entries are engineering-feasible.

Critical note: the winner remains robust inside this scenario under the current weighted scoring contract.

7.4 S4

Algorithm	Status	Weighted score	Task 6 area (m^2)	Max disp. (mm)	Max stress (MPa)	Buckling factor
BFGS	Engineering-feasible	10.800	7779.85	3.831	2.282	20.267
PSO	Warning case	36.300	7931.66	12.073	4.541	22.491
SA	Engineering-feasible	12.100	7769.88	4.287	2.296	19.753

Winner: S4 / BFGS with weighted score `10.800`.

The score gap to second place (S4 / SA) is `1.300` points.

Spread/Risk: displacement spans `3.831` to `12.073` mm, stress spans `2.282` to `4.541` MPa, and buckling spans `19.753` to `22.491`.

Status caveat: warning-only entries are present for PSO and are not decision-grade designs.

Critical note: scenario fairness is reduced by mixed compliance statuses; ranking penalties keep the comparison visible but do not make warning/fallback cases equivalent to engineering-feasible designs.

7.5 S5

Algorithm	Status	Weighted score	Task 6 area (m^2)	Max disp. (mm)	Max stress (MPa)	Buckling factor
BFGS	Engineering-feasible	10.600	7742.56	8.695	4.148	22.725
PSO	Engineering-feasible	18.150	7747.93	11.481	4.941	19.566
SA	Engineering-feasible	13.350	7749.52	10.229	4.642	23.232

Winner: S5 / BFGS with weighted score `10.600`.

The score gap to second place (S5 / SA) is `2.750` points.

Spread/Risk: displacement spans `8.695` to `11.481` mm, stress spans `4.148` to `4.941` MPa, and buckling spans `19.566` to `23.232`.

Status caveat: all three entries are engineering-feasible.

Critical note: the winner remains robust inside this scenario under the current weighted scoring contract.

7.6 S6

Algorithm	Status	Weighted score	Task 6 area (m^2)	Max disp. (mm)	Max stress (MPa)	Buckling factor
BFGS	Engineering-feasible	10.150	7741.72	10.522	5.312	24.772
PSO	Engineering-feasible	16.600	7748.25	11.907	6.705	21.611
SA	Engineering-feasible	21.950	7783.24	14.270	7.690	19.128

Winner: S6 / BFGS with weighted score `10.150`.

The score gap to second place (S6 / PSO) is `6.450` points.

Spread/Risk: displacement spans `10.522` to `14.270` mm, stress spans `5.312` to `7.690` MPa, and buckling spans `19.128` to `24.772`.

Status caveat: all three entries are engineering-feasible.

Critical note: the winner remains robust inside this scenario under the current weighted scoring contract.

7.7 S7

Algorithm	Status	Weighted score	Task 6 area (m^2)	Max disp. (mm)	Max stress (MPa)	Buckling factor
BFGS	Engineering-feasible	6.750	7740.66	10.105	4.134	24.948
PSO	Math fallback	26.250	8324.25	28.467	9.258	23.712
SA	Math fallback	22.350	7890.85	12.541	6.711	26.587

Winner: S7 / BFGS with weighted score `6.750`.

The score gap to second place (S7 / SA) is `15.600` points.

Spread/Risk: displacement spans `10.105` to `28.467` mm, stress spans `4.134` to `9.258` MPa, and buckling spans `23.712` to `26.587`.

Status caveat: mathematical fallback entries are present for PSO, SA.

Critical note: scenario fairness is reduced by mixed compliance statuses; ranking penalties keep the comparison visible but do not make warning/fallback cases equivalent to engineering-feasible designs.

7.8 S8

Algorithm	Status	Weighted score	Task 6 area (m^2)	Max disp. (mm)	Max stress (MPa)	Buckling factor
BFGS	Warning case	39.600	7813.24	10.319	5.446	19.273
PSO	Warning case	27.400	7899.07	9.235	4.493	27.335
SA	Warning case	27.700	7718.27	8.163	4.955	23.837

Winner: S8 / PSO with weighted score `27.400`.

The score gap to second place (S8 / SA) is `0.300` points.

Spread/Risk: displacement spans `8.163` to `10.319` mm, stress spans `4.493` to `5.446` MPa, and buckling spans `19.273` to `27.335`.

Status caveat: warning-only entries are present for BFGS, PSO, SA and are not decision-grade designs.

Critical note: this scenario is tightly clustered, so small modeling shifts can reorder first and second place.

Chapter 8

Convergence Summary

The verification workflow solves both coarse and refined meshes for every case. The acceptance criteria are: displacement change below `5.0%`, stress change below `5.0%`, and first buckling-factor change below `10.0%`.

At the current stage, **0 of 24 comparison pairs** satisfy all three convergence criteria.

Because the study is currently at `0/24` all-pass convergence, the ranking should be interpreted as comparative screening quality, not final structural qualification.

Case	Status	Delta disp. (%)	Pass	Delta stress (%)	Pass	Delta buckling (%)	Pass	All pass
S1 / BFGS	Warning case	8.66	no	3.95	yes	-51.38	no	no
S1 / PSO	Engineering-feasible	17.03	no	8.32	no	-51.39	no	no
S1 / SA	Engineering-feasible	9.59	no	2.24	yes	-49.67	no	no
S2 / BFGS	Engineering-feasible	27.17	no	0.60	yes	-48.05	no	no
S2 / PSO	Math fallback	-11.47	no	-60.11	no	-5.64	yes	no
S2 / SA	Engineering-feasible	25.49	no	-1.19	yes	-46.75	no	no
S3 / BFGS	Engineering-feasible	26.57	no	-0.73	yes	-48.06	no	no
S3 / PSO	Engineering-feasible	27.31	no	1.44	yes	-50.01	no	no
S3 / SA	Engineering-feasible	21.16	no	-3.57	yes	-46.37	no	no
S4 / BFGS	Engineering-feasible	-5.76	no	7.01	no	-51.16	no	no
S4 / PSO	Warning case	16.53	no	-3.15	yes	-158.30	no	no
S4 / SA	Engineering-feasible	-7.69	no	1.43	yes	-49.02	no	no
S5 / BFGS	Engineering-feasible	-5.89	no	10.45	no	-46.31	no	no
S5 / PSO	Engineering-feasible	1.17	yes	23.43	no	-49.25	no	no
S5 / SA	Engineering-feasible	-2.76	yes	5.00	no	-45.85	no	no
S6 / BFGS	Engineering-feasible	2.08	yes	20.49	no	-41.81	no	no
S6 / PSO	Engineering-feasible	17.42	no	39.74	no	-54.85	no	no
S6 / SA	Engineering-feasible	15.36	no	38.65	no	-55.49	no	no
S7 / BFGS	Engineering-feasible	17.31	no	0.69	yes	-43.58	no	no
S7 / PSO	Math fallback	11.87	no	-19.73	no	-240.70	no	no
S7 / SA	Math fallback	-6.17	no	-19.68	no	-63.19	no	no
S8 / BFGS	Warning case	2.71	yes	10.91	no	-53.42	no	no
S8 / PSO	Warning case	7.91	no	-0.58	yes	-49.86	no	no
S8 / SA	Warning case	5.04	no	15.88	no	-53.02	no	no

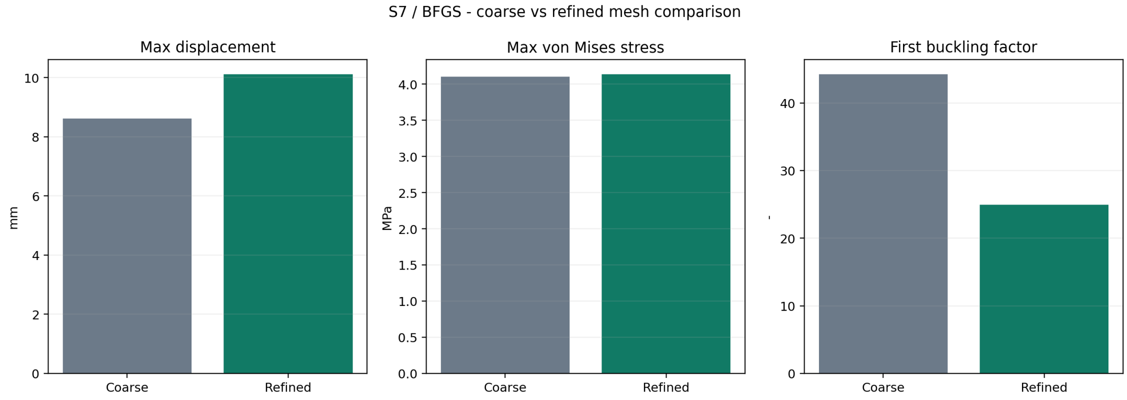

Figure 8.1: Winner mesh comparison

The global winner is therefore reported as **provisional**: the ranking is valid for comparative screening, but convergence evidence is not yet strong enough for a final structural sign-off.

Chapter 9

Warning Cases from Scenario S8

Scenario S8 is kept intentionally as a warning family. None of the Task 6 S8 runs are mathematically compliant or engineering-feasible, so these Abaqus models are useful only as structural cautionary examples and not as candidate towers for recommendation.

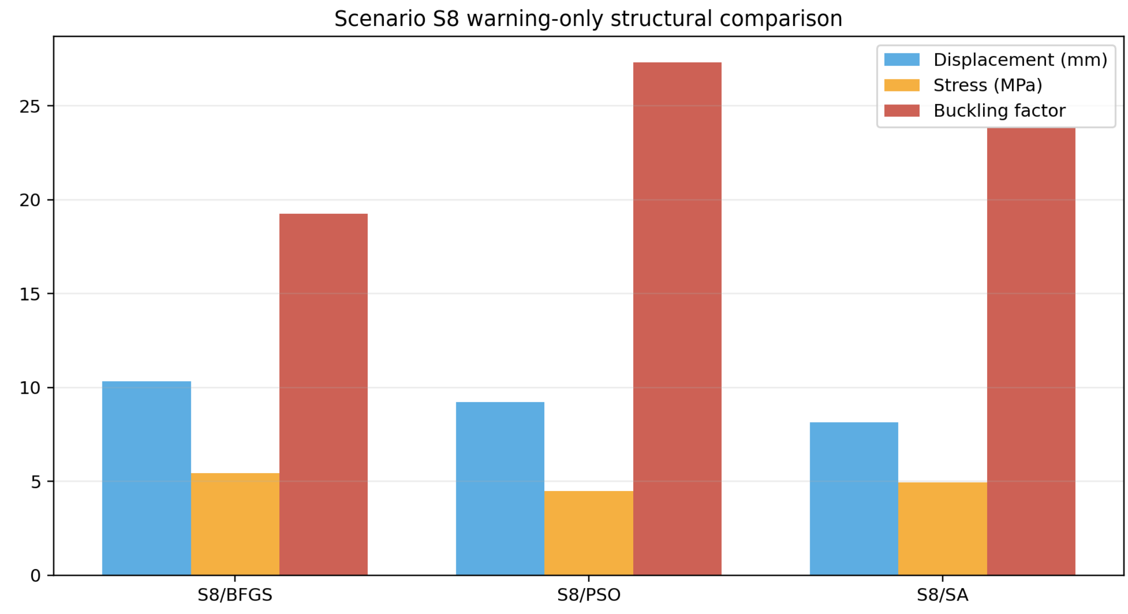

Figure 9.1: Scenario S8 warning metrics

Chapter 10

Field Visualizations of the Global Top-3

All detailed field figures are rendered from actual Abaqus ODB data with Python, not from Abaqus screenshots.

Task 7 figures use a true-scale equal-axis policy (no axis stretching). This differs from some legacy Task 6 renderings that used a different plotting style.

10.1 S7 / BFGS

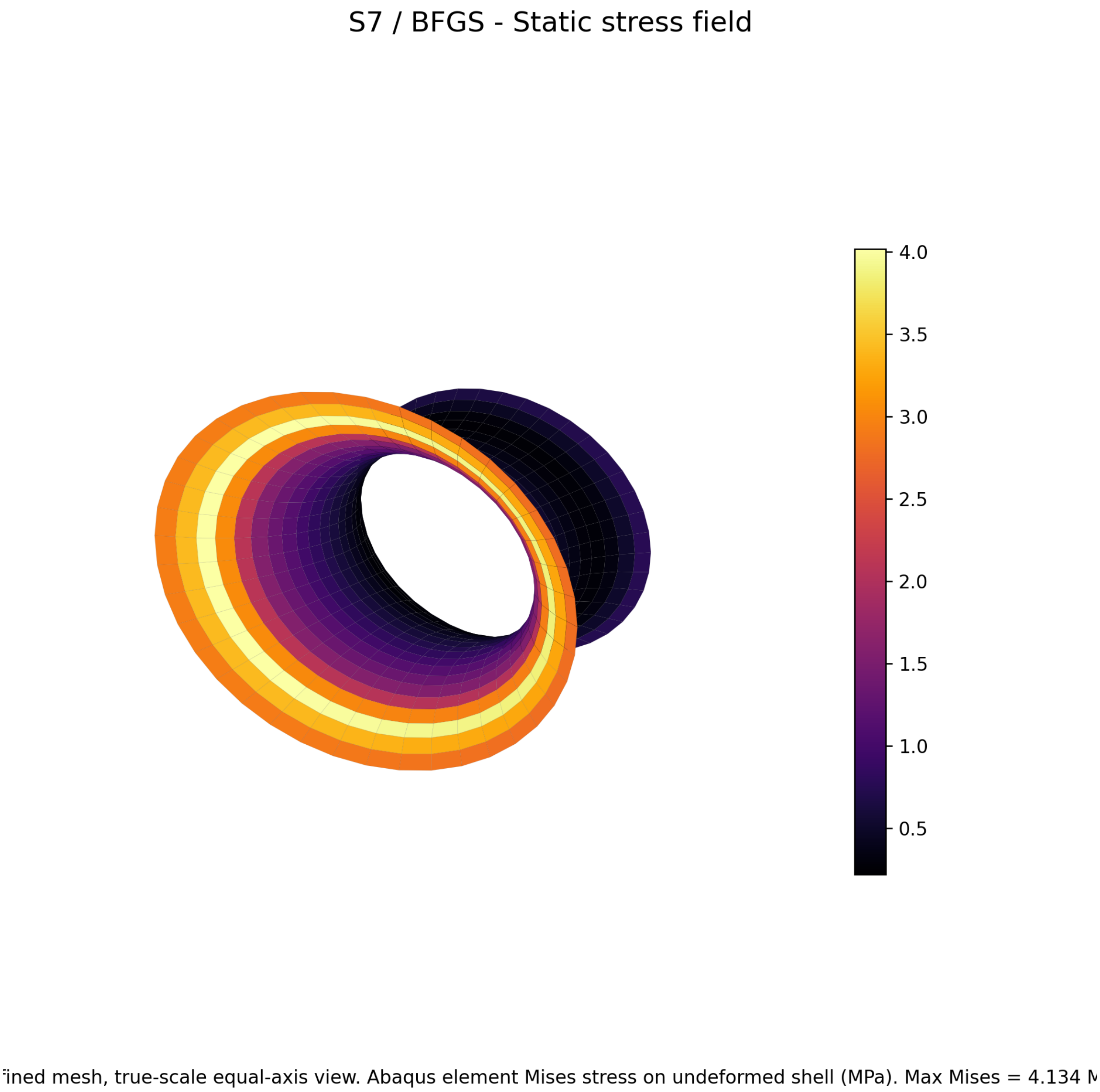

Figure 10.1: S7 / BFGS stress

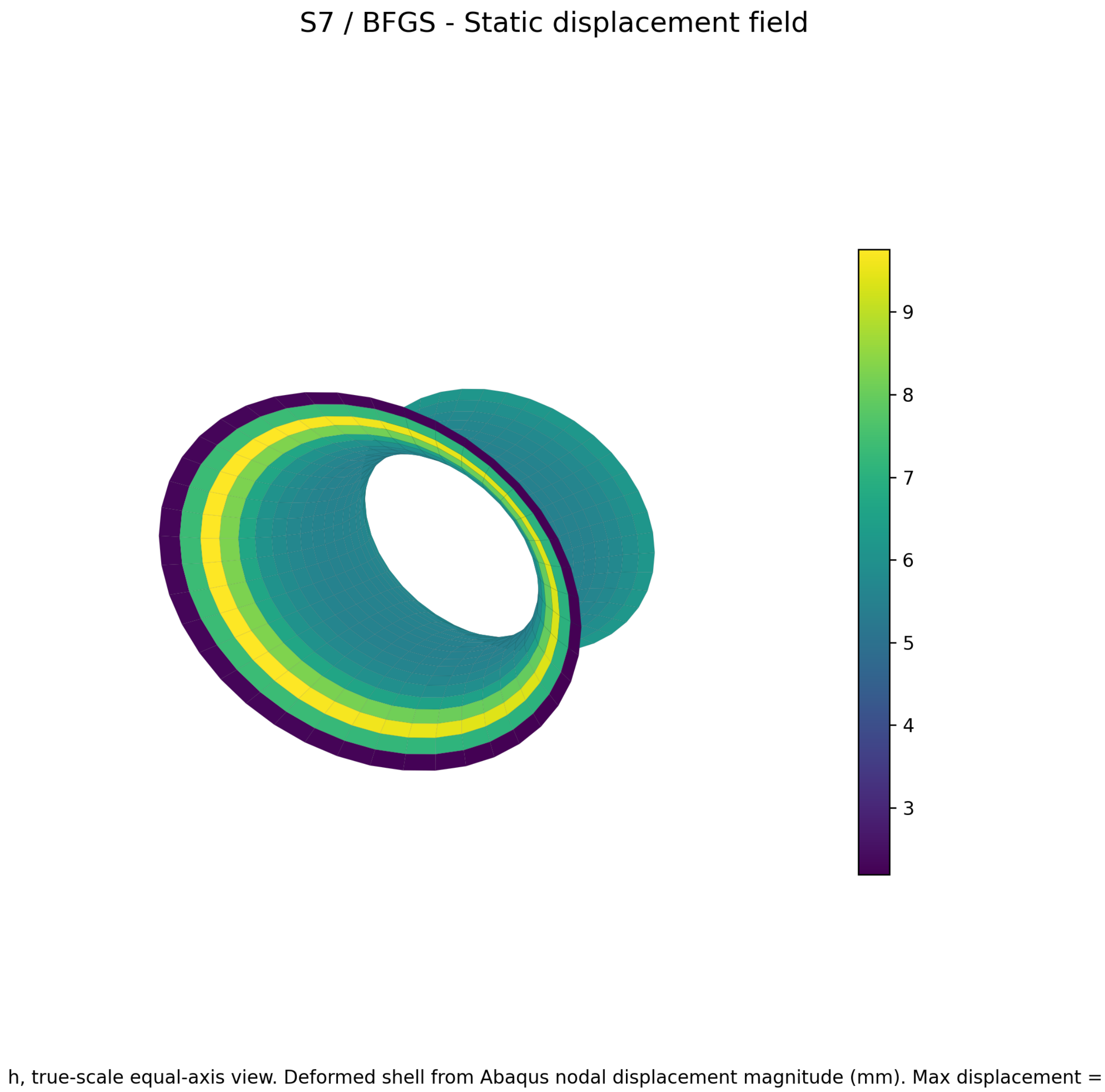

Figure 10.2: S7 / BFGS displacement

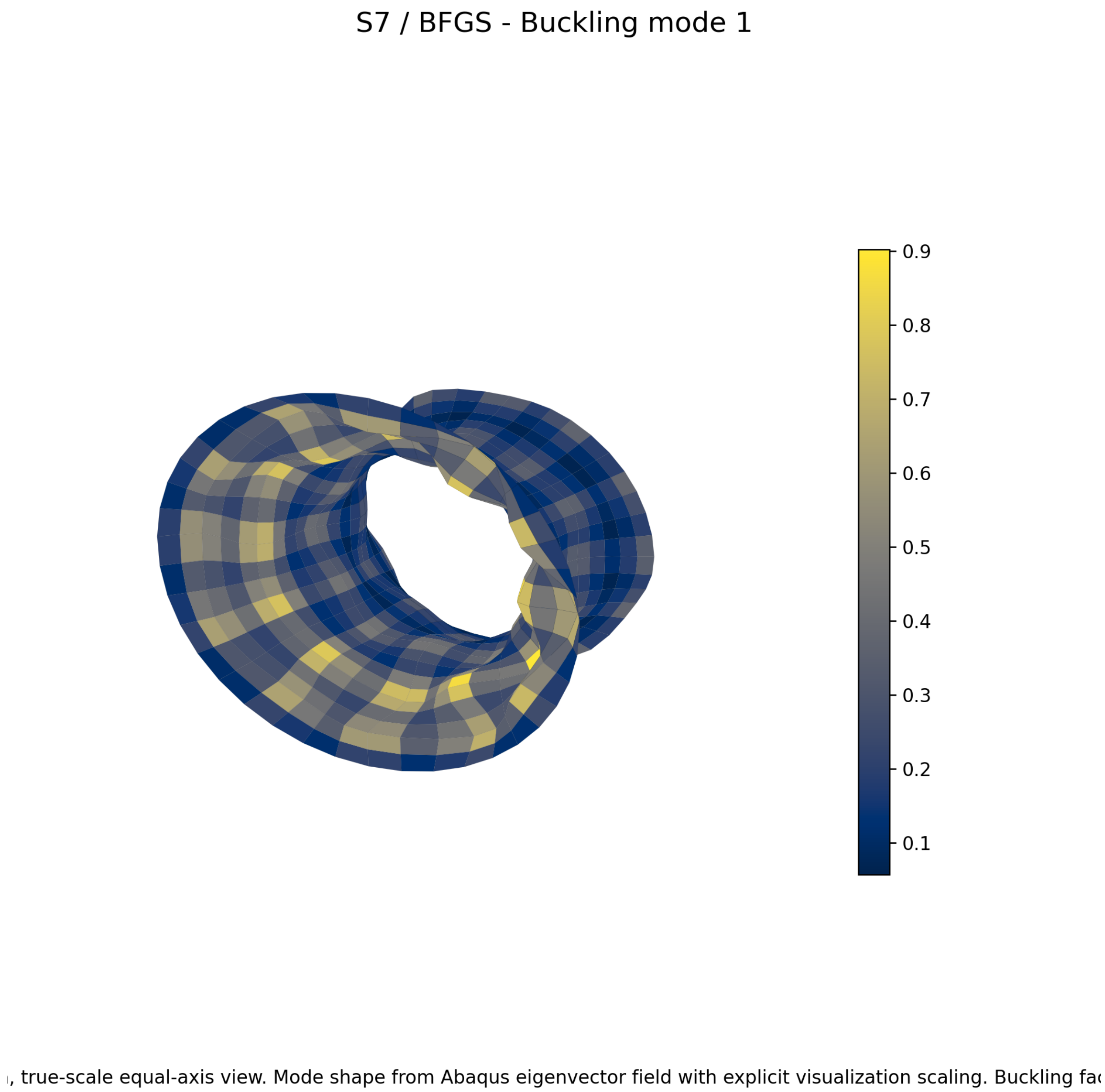

Figure 10.3: S7 / BFGS buckling

10.2 S3 / SA

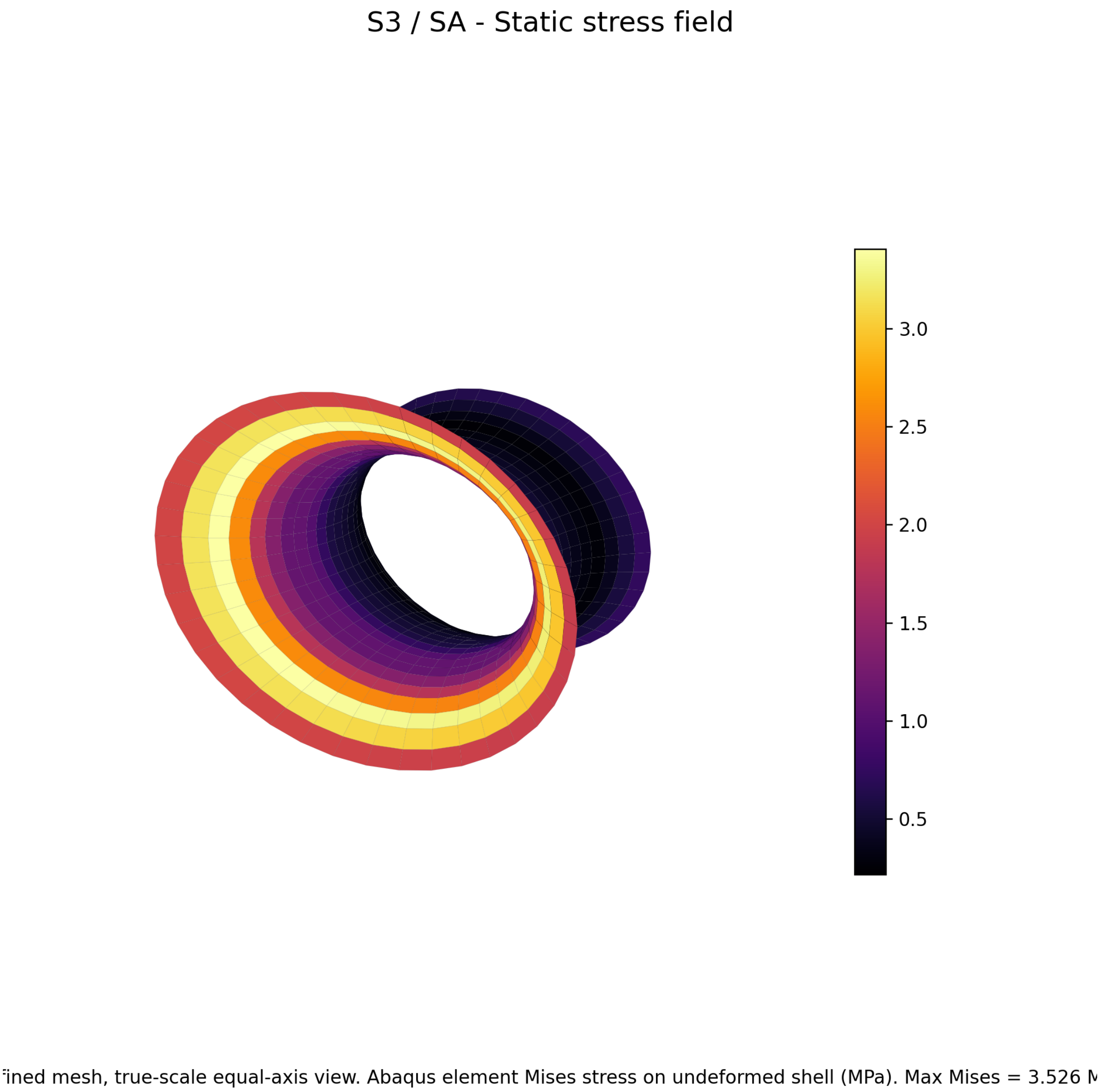

Figure 10.4: S3 / SA stress

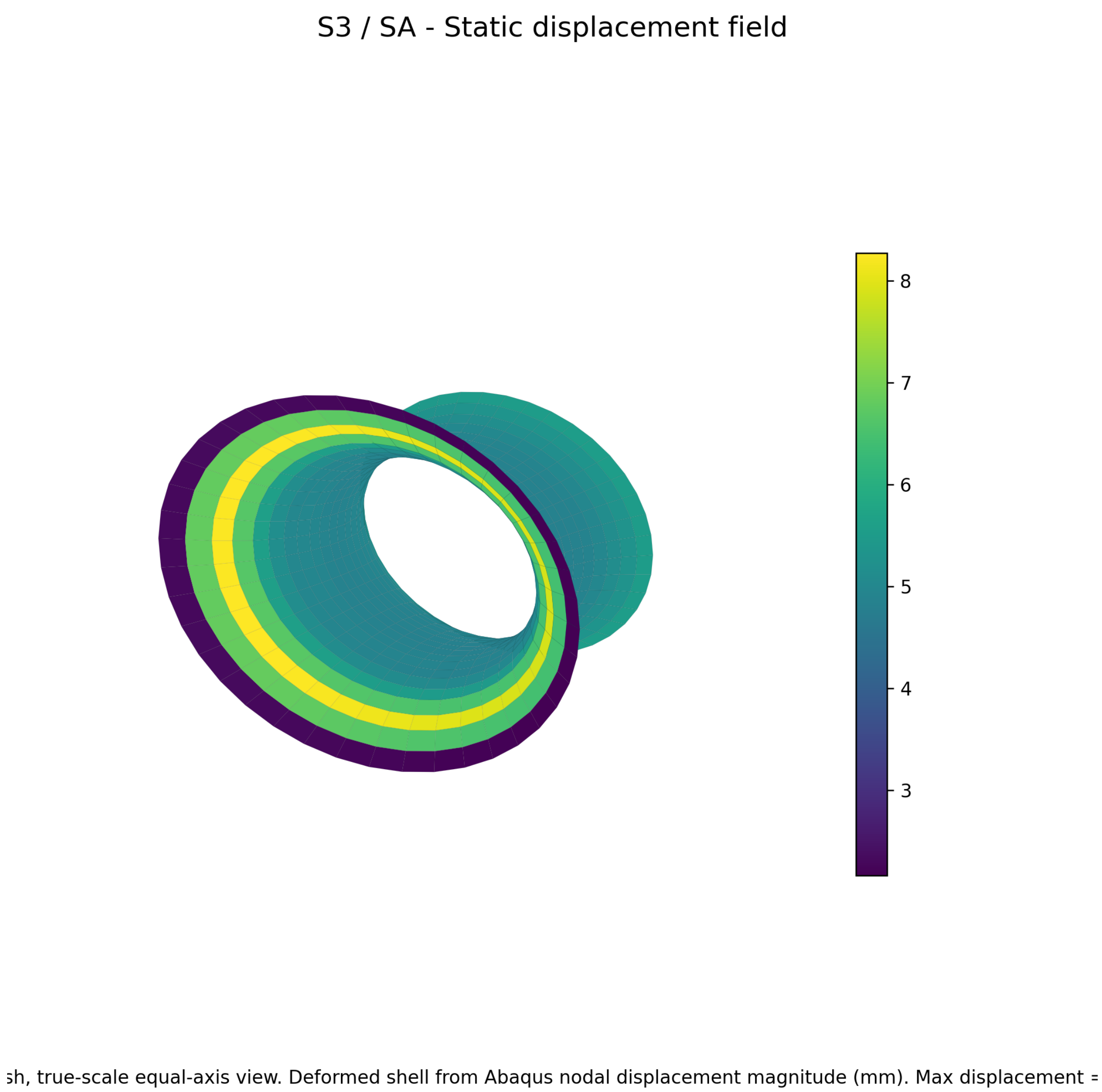

Figure 10.5: S3 / SA displacement

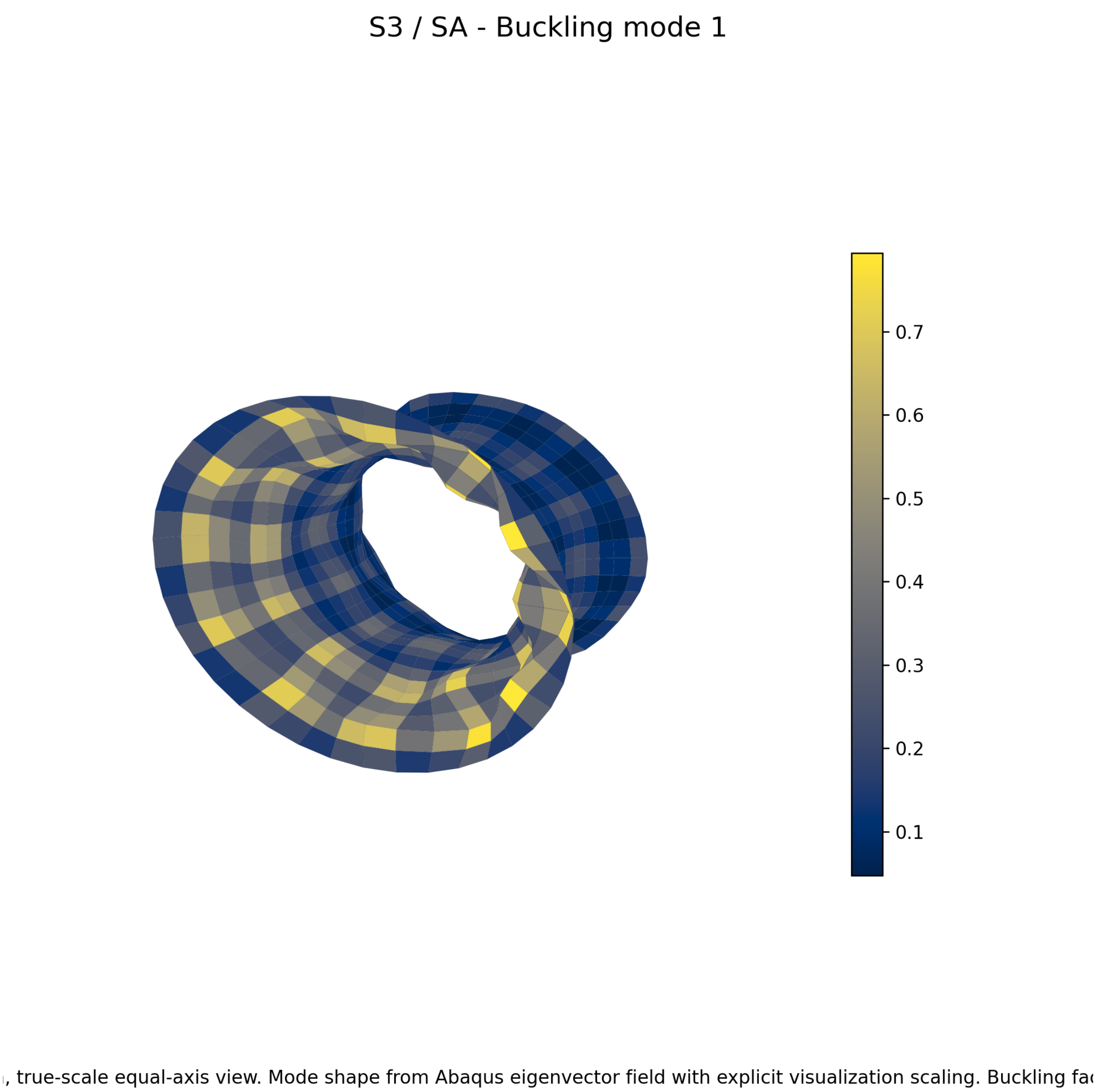

Figure 10.6: S3 / SA buckling

10.3 S2 / SA

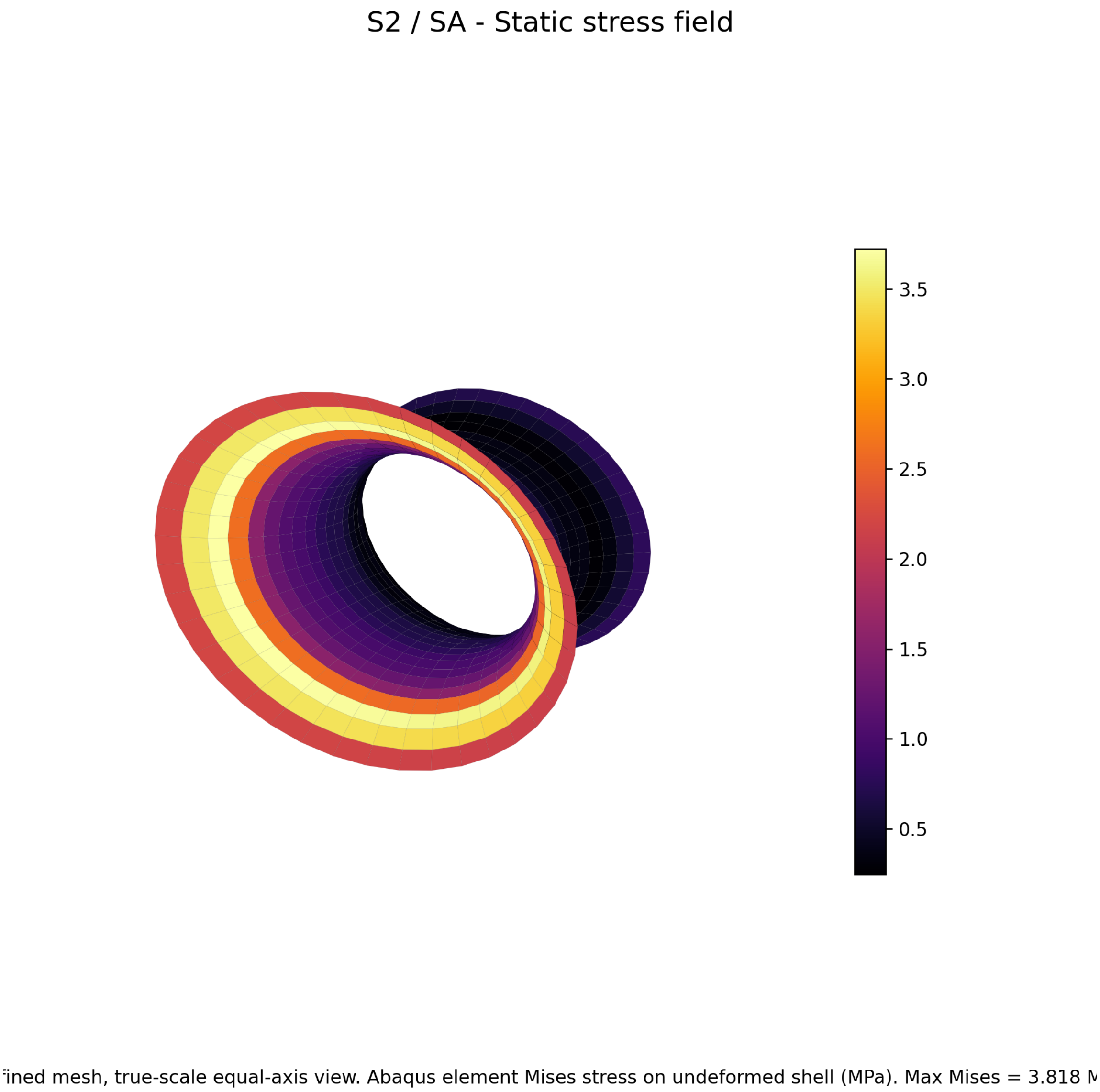

Figure 10.7: S2 / SA stress

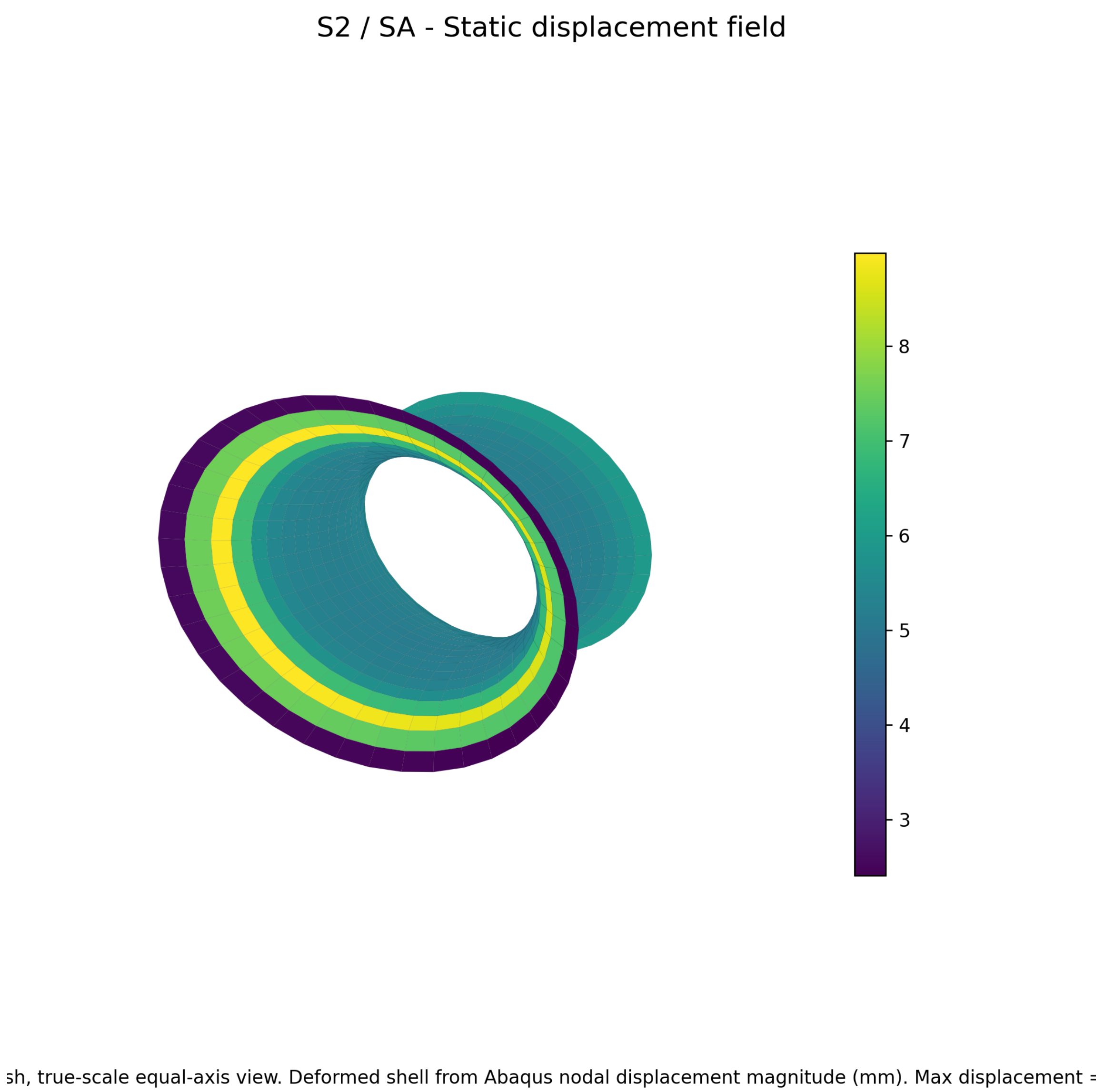

Figure 10.8: S2 / SA displacement

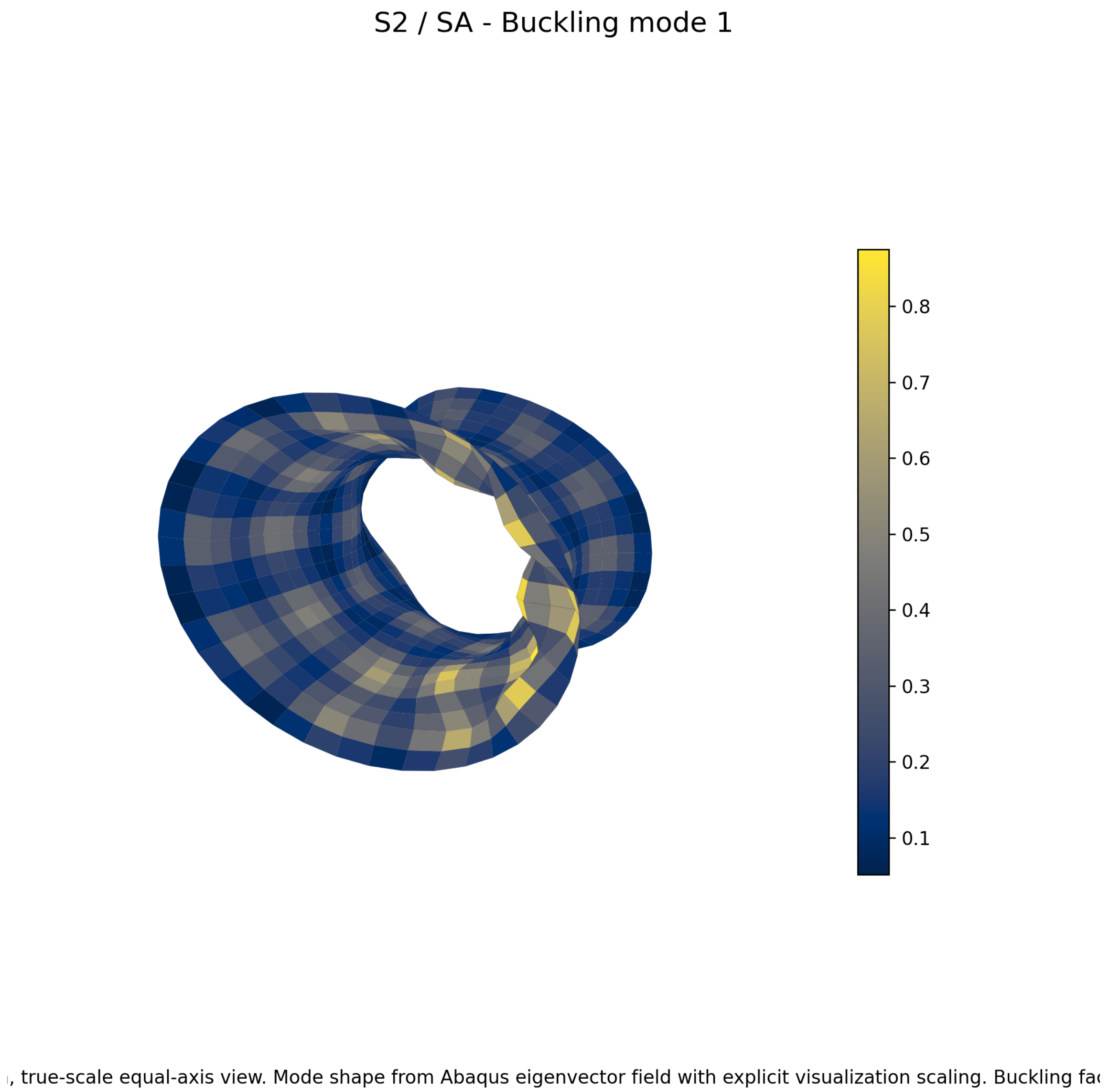

Figure 10.9: S2 / SA buckling

Chapter 11

Recommendation for Later Tasks

The current Task 7 recommendation is **S7 / BFGS** with a `provisional` status. Its weighted score is `6.750` and its refined response gives a first buckling factor of `24.948`, a maximum displacement of `10.105 mm`, and a maximum stress of `4.134 MPa`.

The main practical outcome for the project is that Task 6 geometric alternatives can now be compared with one consistent structural score, explicit warning handling, and scenario-level interpretation suitable for Task 9 communication.

Bibliography

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[3] S. P. Timoshenko and S. Woinowsky-Krieger, *Theory of Plates and Shells*. McGraw-Hill, 1959.